

Inputs

X: queries

	p0	p1	p2	p3
pos 0	1	0	0	0
pos 1	0	1	0	0
pos 2	0	0	1	0
pos 3	0	0	0	1

Y: memory

	s0	s1	s2	s3
pos 0	1	0	0	0
pos 1	0	1	0	0
pos 2	0	0	1	0
pos 3	0	0	0	1
content	3	1	4	2

target (correct answers)

	p0	p1	p2	p3
slot p+1	1	4	2	3
slot p+3	2	3	1	4

task: query at p must return
content of slot p+1 (head 1's job)
content of slot p+3 (head 2's job)

Head 1 (looks 1 slot ahead)

Linear: weights x inputs

$W_q (d \times 4)$

0	0	0	12
12	0	0	0
0	12	0	0
0	0	12	0

$Q = W_q X (d \times 4)$

p0	p1	p2	p3
0	0	0	12
12	0	0	0
0	12	0	0
0	0	12	0

$W_k (d \times 5)$

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0

$K = W_k Y (d \times 4)$

s0	s1	s2	s3
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

$W_v (1 \times 5)$

0	0	0	0	1
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$V = W_v Y (1 \times 4)$

s0	s1	s2	s3
3	1	4	2

W_q moves position p to p+1, so Q for query p points at slot p+1

Scaled dot product (L-shape: right factor above, left factor to the left)

$K^T (4 \times d)$

s0	s1	s2	s3
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

Q

p0	p1	p2	p3
0	0	0	12
12	0	0	0
0	12	0	0
0	0	12	0

$S = K^T Q / \sqrt{d}$

0	0	0	6
6	0	0	0
0	6	0	0
0	0	6	0

softmax (columns sum to 1)

p0	p1	p2	p3
0.002	0.002	0.002	0.993
0.993	0.002	0.002	0.002
0.002	0.993	0.002	0.002
0.002	0.002	0.993	0.002

$V (1 \times 4)$

3	1	4	2
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head 1 output $O = V \times \text{softmax}$

1.01	3.99	2	3
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the stripe: each query puts almost all its weight on one slot, so O = the content there

Head 2 (looks 3 slots ahead)

Linear: weights x inputs

$W_q (d \times 4)$

0	12	0	0
0	0	12	0
0	0	0	12
12	0	0	0

$Q = W_q X (d \times 4)$

p0	p1	p2	p3
0	12	0	0
0	0	12	0
0	0	0	12
12	0	0	0

$W_k (d \times 5)$

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0

$K = W_k Y (d \times 4)$

s0	s1	s2	s3
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

$W_v (1 \times 5)$

0	0	0	0	1
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$V = W_v Y (1 \times 4)$

s0	s1	s2	s3
3	1	4	2

W_q moves position p to p+3, so Q for query p points at slot p+3

Scaled dot product (L-shape: right factor above, left factor to the left)

$K^T (4 \times d)$

s0	s1	s2	s3
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

Q

p0	p1	p2	p3
0	12	0	0
0	0	12	0
0	0	0	12
12	0	0	0

$S = K^T Q / \sqrt{d}$

0	6	0	0
0	0	6	0
0	0	0	6
6	0	0	0

softmax (columns sum to 1)

p0	p1	p2	p3
0.002	0.993	0.002	0.002
0.002	0.002	0.993	0.002
0.002	0.002	0.002	0.993
0.993	0.002	0.002	0.002

$V (1 \times 4)$

3	1	4	2
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head 2 output $O = V \times \text{softmax}$

2	3	1.01	3.99
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the stripe: each query puts almost all its weight on one slot, so O = the content there

Combine: each head's output x its valve g_h (supply) x its output weight

both valves open, g = (1, 1)

	p0	p1	p2	p3
head 1	1.01	3.99	2	3
head 2	2	3	1.01	3.99

valve 2 closed, g = (1, 0)

	p0	p1	p2	p3
	1.01	3.99	2	3
	0	0	0	0

target

	p0	p1	p2	p3
	1	4	2	3
	2	3	1	4

with both valves open the output matches the target;
closing valve 2 removes head 2's row exactly